SIMPLE AND COMPOUND INTEREST

MATH035
Mathematics in the Modern World



OBJECTIVE:

To define different terms and be able to solve problems involving simple interest and compound interest.





TYPES OF INTEREST

Simple Interest –an amount added in which only the original principal bears interest for the entire term.



FORMULAS:

Simple Interest: I = P r t

P - principal

r – annual interest rate

t – term or time (years)



Future Value: F = P + I

TERMS OF LOAN

1) When the time is expressed in number of year(s), our formula will be:

$I = (P)(r)(\text{number of years})$



TERMS OF LOAN

2) When time is expressed in number of month(s):

$$I=(P)(r)\left(\frac{\text{number of months}}{12}\right)$$



TERMS OF LOAN

3) When time is expressed in number of day(s):

- Ordinary Interest (I_o)

  $I_o = (P)(r)\ \frac{\text{(number of days)}}{360}$

- Exact Interest (I_e)

  $I_e = (P)(r)\ \frac{\text{(number of days)}}{365}$



PROBLEM 1

Find the interest and future value of Php900 at 7.25% simple interest for 9 months.

Given: P=Php900, r=0.0725, t=9months=9/12 year=3/4 year

Find: I and F

Solution: I=Prt=(900)(0.0725)(3/4)=Php48.94

F=P+I=900+48.94=Php948.94



PROBLEM 2

Find the interest and future value of Php1,200 at 11 3/8% simple interest for 2 years and 5 months.



Given: P=Php1200, r= 11 3/8% =11.375%=0.11375, t=2yrs and 5months=2 + 5/12 years=29/12 years

Find: I and F

Solution: I=Prt=(1200)(0.11375)(29/12)=Php329.88

F=P+I=1200+329.88=Php1,529.88

PROBLEM 3

If a principal of Php2,500 earns interest of Php185 in 3 years and 4 months, what interest rate is in effect?



Given: P=Php2500, I=Php185,
t=3yrs and 4months= 3 + 4/12 =10/3 years

Find: r

Solution: I=Prt

r=I / (Pt)=185 / ((2500)(10/3))=0.0222=2.22%

PROBLEM 4

A principal earns interest of Php385 in 2 years and 9 months at a simple interest rate of 9 ½%. Find the principal invested.



Given: I=Php385, r=9 ½%=9.5%=0.095
t=2yrs and 9months= 2 + 9/12 =11/4 years

Find: P

Solution: I=Prt

P=I / (rt)=385 / ((0.095)(11/4))=Php1473.68

PROBLEM 5

Find the exact interest and future value of Php8,800 for 85 days at 11% simple interest.

Given: P=Php8800, r=11%=0.11, t=85 days

Find: I_e and F

Solution:

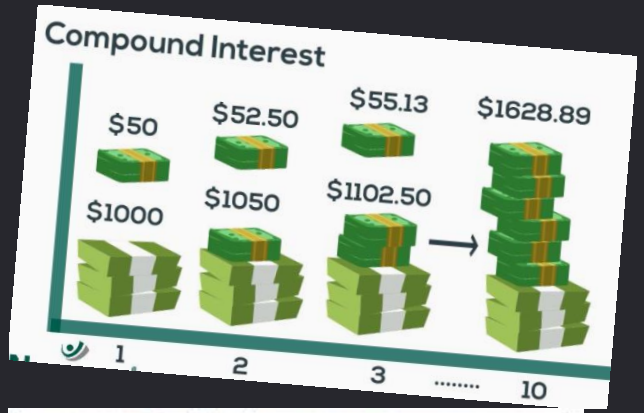
I_e=Prt=(8800)(0.11)(85/365)=Php225.42

F=P+I=8800+225.42=Php9025.42



TYPES OF INTEREST

Compound Interest – an amount added to the principal at the end of a certain period of time, after which the interest is computed on the new principal.

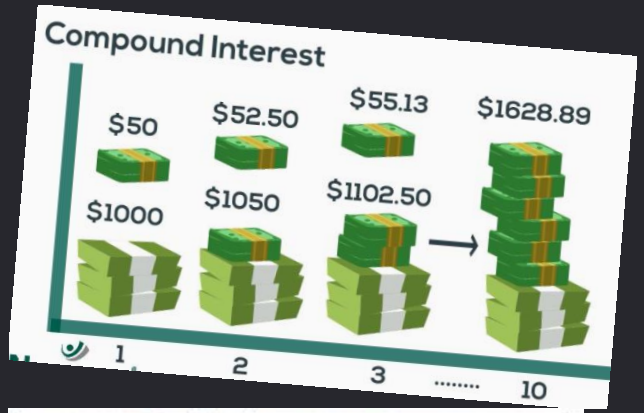




FORMULA

$$F = P(1 + r/m)^{tm}$$

$$I = F - P$$





where:

F = future value or compounded amount

I = interest

P = original principal

r = annual interest rate

t = time in years

m = number of times compounded per year

Note: m=1 if compounded annually, m=2-semi-annually, m=4-quarterly and m=12-monthly.

PROBLEM 1

Php5,500 is invested at 8% compounded quarterly for 5 years and 6 months. Find the interest and the future value of Php5,500.



Given: P=Php5500, r=8%=0.08, m=4(quarterly)
t=5 years and 6 months= 5 + 6/12= 11/2 years

Find: I and F



PROBLEM 1 (cont'd)

Given: P=Php5500, r=8%=0.08, m=4(quarterly)
t=5 years and 6 months= 5 + 6/12= 11/2 years

Find: I and F

Solution: $F = P(1 + r/m)^{tm}$

$F = 5500(1 + 0.08/4)^{(11/2)(4)} =$ Php8502.89

$I = F - P = 8502.89\text{-}5500 =$ Php3002.89





PROBLEM 2

How long (in years) will it take Php4, 500 to amount to Php6,100 if the interest rate is 5% compounded monthly?

Given: P=Php4500, F=Php6100, r=5%=0.05
m=12 (monthly)

Find: t





PROBLEM 2 (cont'd)

Given:

P=Php4500

F=Php6100,

r=5%=0.05

m=12 (monthly)

Find: t

Solution:

Step No.	Steps	Explanation
1.	$F = P\left(1 + \frac{r}{m}\right)^{tm}$	
2.	$6100 = 4500\left(1 + \frac{0.05}{12}\right)^{t(12)}$	Substitute the given.
3.	$6100 = 4500\left(\frac{241}{240}\right)^{12t}$	Add: $1 + \frac{0.05}{12}$
4.	$\frac{6100}{4500} = \left(\frac{241}{240}\right)^{12t}$	Divide both sides by 4500
5.	$\ln\left(\frac{6100}{4500}\right) = \ln\left(\frac{241}{240}\right)^{12t}$	Take the natural logarithm (ln) of each side.
6.	$\ln\left(\frac{6100}{4500}\right) = 12t \ln\left(\frac{241}{240}\right)$	Property of Logarithm: $\ln b^x = x \ln b$
7.	$\frac{\ln\left(\frac{6100}{4500}\right)}{12 \ln\left(\frac{241}{240}\right)} = t$	Divide both sides by $12 \ln\left(\frac{241}{240}\right)$
8.	$t = 6.10\ years$	Simplify.

PROBLEM 3

Find the interest rate if the compounded amount is Php21,000 and the invested capital Php15,400 for 5 years and 8 months compounded semi-annually.





Given: P=Php15400, F=Php21000, t=5years and 8 months= 5 + 8/12 =17/3 years, m=2 (semi-annually)

Find: r

PROBLEM 3 (cont'd)

Given:

P=Php15400

F=Php21000,

t=5years and 8 months

= 5 + 8/12 =17/3 years

m=2 (semi-annually)

Find: r

Solution:

Step No.	Steps	Explanation
1.	$F = P\left(1+\frac{r}{m}\right)^{tm}$	
2.	$21000 = 15400\left(1+\frac{r}{2}\right)^{\left(\frac{17}{3}\right)(2)}$	Substitute the given.
3.	$\frac{21000}{15400} = \left(1+\frac{r}{2}\right)^{\frac{34}{3}}$	Divide both sides by 15400. Multiply: $\left(\frac{17}{3}\right)(2)$
4.	$\left(\frac{21000}{15400}\right)^{\frac{3}{34}} = \left(\left(1+\frac{r}{2}\right)^{\frac{34}{3}}\right)^{\frac{3}{34}}$	Raise both sides to $\frac{3}{34}$
5.	$\left(\frac{21000}{15400}\right)^{\frac{3}{34}} = 1+\frac{r}{2}$	Law of exponent: $(x^a)^b=x^{ab}$
6.	$\left(\frac{21000}{15400}\right)^{\frac{3}{34}} - 1 = \frac{r}{2}$	Transpose 1 from the right to the left hand side.
7.	$2\left[\left(\frac{21000}{15400}\right)^{\frac{3}{34}} - 1\right] = r$	Multiply both sides by 2
8.	$r = 0.0555 \text{ or } 5.55\%$	Simplify.